Org Activity Tracker

What it does, what it can do today, and what's still missing

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1. What This Project Does

An internal employee time & activity tracker for the organization — the same kind of thing Hubstaff or the internal DeskQ tool does, built in-house instead of paid per seat. An employee installs a small app that sits in their system tray. They click **Start Tracking**, and from that moment the system records their session, takes a screenshot of their screen automatically every 5 minutes, and logs the machine's IP address with each one. They click **Stop Tracking** when done. A manager logs into a separate web dashboard to see who is tracking right now, how long each session lasted, and the screenshots captured during it.

It's built from three parts that only talk to each other over the network: a central backend (the database and all the logic), a desktop agent (the tray app on each employee's PC), and an admin dashboard (the manager's web view). One backend serves every employee's agent and the dashboard at once.

2. Functionalities — What Works Today

2.1 For the employee (desktop agent)

- Logs in once through a small popup login window (no terminal/typing commands needed) — the session is remembered afterward so it doesn't ask again until it naturally expires.
- System tray icon with **Start Tracking** / **Stop Tracking** / **Quit** — the icon itself turns green while actively tracking, gray when stopped.
- Automatic screenshot every 5 minutes (adjustable) while tracking is on, tagged with the current IP address and timestamp.
- Popup notifications for every important moment: a screenshot was taken, tracking started or stopped, or something went wrong (server unreachable, upload failed, login expired) — so the employee always knows the current state even with no window open.
- If the app or PC restarts mid-session, it automatically reconnects to the still-running session on the server instead of losing track of it.
- Packaged as a single Windows program (no Python installation needed on employee machines) that can be set to launch automatically every time the employee logs into Windows.

2.2 For the manager/admin (web dashboard)

- Secure login, with two account types: admin (sees everyone) and employee (sees only their own activity, if they were ever given dashboard access).
- **Team Overview** page — see at a glance who is actively tracking right now and the IP their current session is coming from.
- **Timesheets** page — full history of every session per employee: start time, end time, total duration, filterable by person, all shown correctly in the local time zone.
- **Screenshots** page — gallery of every captured screenshot per session, with a click-to-enlarge full-screen viewer (click outside, press the X, or hit Escape to close).

2.3 Underneath (backend & data)

- One central database holding every employee, every session, and every screenshot with its IP address and timestamp — the single source of truth for the whole organization.
- Passwords are hashed before storage (bcrypt) — never stored or ever sent back in readable form.
- Ready to scale from one person testing (a zero-setup local database) to the whole company writing at once (swap one setting to use a proper multi-user database).
- Access control so the API can be safely exposed beyond one laptop (restricting which websites/dashboards are allowed to talk to it) once it's centrally hosted.

All of the above has been built and exercised end-to-end (login → start tracking → screenshot capture → stop tracking → viewing it all on the dashboard). The screenshot capture and login popup's on-screen appearance still need to be confirmed on a real Windows machine, since this was built in an environment without one.

3. What's Missing / Not Built Yet

Nothing below is a secret or an oversight — each is a deliberate next step, ranked by how much it matters before this touches real coworkers' machines.

Gap	Why it matters	Priority
Employee notice & consent process	Screen capture + IP logging of staff is legally regulated in most places (several US states, EU/GDPR-influenced regions, India's IT Act all expect some form of notice). This is a policy step, not code — but it blocks everything else.	Must do before rollout
"Add Employee" screen in the dashboard	New accounts currently have to be created through the raw API documentation page instead of a proper form.	Medium
Retry/queue for failed screenshot uploads	If the network hiccups right when a screenshot is due, that interval's screenshot is simply lost — not retried or saved for later.	Medium
HTTPS in front of the backend	Right now traffic between agents/dashboard and the backend isn't encrypted — fine for local testing, not fine once it's running across a real office network.	High, before company-wide use
Code-signing the Windows program	Windows will show an "unrecognized publisher" security warning the first time each employee runs it, since it isn't signed with a certificate.	Medium
Auto-start on macOS/Ubuntu	The tracker itself already runs on macOS and Ubuntu — only the "launch automatically on login" piece has been built for Windows so far.	Low unless targeting Mac/Linux staff
Backend running as an always-on service	Currently started manually and stops if that window is closed — for real 24/7 use it should run as a proper background service on a server that's always on.	High, before company-wide use
Idle-time detection	No automatic pause when an employee steps away — a screenshot will still fire on schedule even during idle time.	Low

Gap	Why it matters	Priority
Screenshot redaction/blurring	No automatic blurring of sensitive on-screen info (passwords, personal chats) before a screenshot is stored.	Low, but relevant to the privacy discussion above
Password reset / self-service account tools	An admin currently has to handle any account issue manually — no "forgot password" flow.	Low

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For the full technical breakdown (every file, why each technology was chosen, and how data flows end-to-end) see the companion document: Org-Tracker-Code-Walkthrough.pdf.